

Xiaotian XU

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4050 Natural History Building, 1301 W Green St, Urbana, IL, 61801

RESEARCH INTEREST

Aerosol Modeling, Aerosol-Cloud Interactions, Aerosol-Climate Impacts

PROFESSIONAL EXPERIENCES

Postdoctoral Associate

University at Albany, State University of New York

starting Aug. 2026

Advisor: Prof. **Fangqun Yu**

Graduate Teaching Assistant

University of Illinois Urbana-Champaign

Sep. 2024 - Dec. 2024, Jan. 2026 - Jul. 2026

Graduate Research Assistant

University of Illinois Urbana-Champaign

Aug. 2021 - Aug. 2024, Jan. 2025 - Dec. 2025

Advisor: Prof. **Nicole RIEMER**

Research Assistant

Nanjing University, School of Atmospheric Science

Dec. 2019 – Jun. 2021

Advisor: Prof. **Yanxu ZHANG**

EDUCATION

University of Illinois Urbana-Champaign

Ph.D.: Atmospheric Science

Aug. 2021 –

Advisor: Prof. **Nicole RIEMER**

The Hong Kong University of Science and Technology

M.Sc.: Environmental Science & Management

Sep. 2018 – Nov. 2019

Advisor: Prof. **Jianzhen YU**

Shanxi University

B.Eng.: Environmental Engineering

Sep. 2014 – Jun. 2018

PUBLICATIONS

In prep & under review

12. Shen, W., Wang, J., Riemer, N., West, M., Wang, M.*, **Xu, X.***, Jiang, F., and Zheng, Z., “Machine learning-derived surface tension in climate models and its implications for CCN activation”, in prep
11. Connolly, P.*, Bi, K., Chen, S., Deng, Z., Etcheto, J., Hu, Y., Kim, N., Riemer, N., Tang, W., Topping, D., Umo, N. S., Wagner, R., **Xu, X.**, Xue, L., Yum, S. S., Zhu, B., and Möhler, O., “Investigating the competition in cloud droplets activation by different aerosol systems using cloud chamber experiment datasets and modelling”, in prep
10. **Xu, X.**, Jiang, F., Curtis, J. H., Zheng, Z., West, M., and Riemer, N.*, “A diagnostic scaling framework for connecting surfactant-induced CCN perturbations to global aerosol-cloud forcing patterns”, in prep
9. Jiang, F., Coe, H., Chan, A. W. H., Connolly, P., Liu, D., Li, W., Zhang, Y., Liu, X., Lee, A. K. Y., Zhang, Y., Zhang, C., Riemer, N., **Xu, X.**, Yu, J., Yu, J., Ching, J.*, and Zheng, Z.*, “Nanoplastics are a small but emerging perturbation to global warm clouds”, in prep
8. **Xu, X.**, Rivera-Adorno, F. A., Laskin, A.*, West, M., and Riemer, N.*, “Diagnosing the Role of Mixing State in Biomass-Burning Aerosol Optical Properties Using Particle-Resolved Modeling”, submitted to *Environmental Science: Atmospheres*

Peer reviewed

7. Wang, J., Shen, W.*, Wang, M., Zheng, Z., Jiang, F., Li, T., Liu, Y., Yue, M., Dong, X., Zhu, H., Shao, X., Zhan, C., and **Xu, X.**, “Quantifying Black Carbon Mixing State Heterogeneity Using a Machine Learning Model”, *Journal of Geophysical Research: Atmospheres*, accepted

6. **Xu, X.**, Curtis, J. H., West, M., and Riemer, N.*, “Compensating biases in CCN predictions from composition averaging and neglected surfactant effects”, *Atmospheric Chemistry and Physics*, 2026. DOI: 10.5194/acp-26-10647-2026.
5. **Xu, X.**, Yao, Y., Liu, Y., Curtis, J., West, M., and Riemer, N.*, “Role of liquid-liquid phase separation-induced surface tension changes in cloud droplet activation”, *Aerosol Science and Technology*, 2026. DOI: 10.1080/02786826.2026.2631789.
4. **Xu, X.**, Feng, X., Lin, H., Zhang, P., Huang, S., Song, Z., Peng, Y., Fu, T.-M., and Zhang, Y.*, “Modeling the high-mercury wet deposition in the southeastern US with WRF-GC-Hg v1.0”, *Geoscientific Model Development*, 2022. DOI: 10.5194/gmd-15-3845-2022.
3. Zhao, Y., Zhang, K.*, **Xu, X.**, Shen, H.*, Zhu, X., Zhang, Y., Hu, Y., and Shen, G., “Substantial changes in Nitrogen Dioxide and Ozone after excluding meteorological impacts during the COVID-19 Outbreak in mainland China”, *Environmental Science & Technology Letters*, 2020. DOI: 10.1021/acs.estlett.0c00304.
2. Geng, H.*, Jin, C., Zhang, D., Wang, S., **Xu, X.**, Wang, X., Zhang, Y., Wu, L., and Ro, C.-U.*, “Characterization of Size-resolved Urban Haze Particles Collected in Summer and Winter at Taiyuan city, China Using Quantitative Electron Probe X-Ray Microanalysis”, *Atmospheric Research*, 2017. DOI: 10.1016/j.atmosres.2017.02.005.
1. Zhang, Y., Geng, H.*, Zhang, D., **Xu, X.**, Wang, S., Wang, X., Wang, D., and Ro, C.-U., “Influences of Biomass and Coal Burning on Ambient Fine Particulate Matter over Taiyuan City”, *Journal of Anhui Agriculture Science*, 2016. DOI: 10.13989/j.cnki.0517-6611.2016.05.041.

CONFERENCE PRESENTATIONS

6. **Oral**, From particle-scale surfactant effects to global aerosol-cloud forcing: a machine-learning diagnostic framework, American Geophysical Union, San Francisco, CA, Dec 2026
5. **Poster**, Surfactant Effects on CCN Activation: Particle-Resolved Modeling, Regional Evaluation, and Global Implications, American Meteorological Society, Houston, TX, Jan 2026
4. **Poster**, Impact of surfactants on cloud condensation nuclei activity: from microscale to global scale, Gordon Research Conferences: Atmospheric Chemistry, Newry, ME, Aug 2025
3. **Oral**, Investigating impact of surfactants on cloud condensation nuclei activity with a particle-resolved aerosol model, International Aerosol Modeling Algorithms Conference, Davis, CA, Dec 2023
2. **Poster**, Quantifying the impact of organic films on cloud condensation nuclei activity using a particle-resolved model, American Chemical Society Spring Meeting, Indianapolis, IN, Mar 2023
1. **Poster**, Quantifying Impact of Organic Films on Cloud Condensation Nuclei Activity, Midwest Student Conference for Atmospheric Research, Urbana, IL, Oct 2022

SELECTED AWARDS

Graduate College's Conference Presentation Award	Fall 2025
Atmospheric Chemistry Gordon Research Conference	2025
Samuel, Berle D. and Jeffrey A. Schlesinger Travel Grant	Fall 2023
Samuel, Berle D. and Jeffrey A. Schlesinger Travel Grant	Spring 2023
The Best ES&T Letters Papers in 2020 (DOI: 10.1021/acs.estlett.1c00562)	2021
Independent Research Project Funding, Nanjing University	2019
First Prize on Publication, 15th Shanxi Challenge Cup	2017
Poster Award, 2016 China-Canada Analytical Chemistry Conference	2016
Excellence Prize, National Undergraduate Innovation & Entrepreneurship Training	2016

TEACHING & MENTORING EXPERIENCES

ATMS/ESE 140: Climate and Global Change	Spring 2026
ATMS 301: Atmospheric Thermodynamics	Fall 2024
Graduate-Graduate Mentorship Program	2023 - 2024
Graduate-Undergraduate Mentorship Program	2022 - 2023

OUTREACH ACTIVITIES

Department Affairs Committee (Liaison for International students)	2024 - 2025
AAAR at UIUC Student Chapter	2022 - present
Engineering Open House: "Care for Air"	2023

PROFESSIONAL SERVICE

Journal Reviewer: Atmospheric Chemistry and Physics (ACP), Journal of Geophysical Research: Atmospheres (JGR: Atmospheres), Journal of Hazardous Materials (J. Hazard. Mater.), Scientific Reports (Sci. Rep.)

TECHNICAL SKILLS

Programing Languages: FORTRAN, Python, NCL, shell, CDO, NCO, Git

Model: PartMC-MOSAIC, CESM, GEOS-Chem, WRF-CMAQ, WRF-GC